

# Coachline CX-32 Intercity Bus

## Operator & Fleet Maintenance Manual

Sample document: this is a fictional vehicle created to demonstrate multi-model retrieval in the Fleet Assistance Copilot. Values are realistic but not taken from any real OEM.

### Document Control

| Field            | Value                                                  |
|------------------|--------------------------------------------------------|
| Document ID      | VM-CX32-OM                                             |
| Vehicle model    | CX-32                                                  |
| Variants covered | CX-32 AC Seater (45+1+1), CX-32 AC Sleeper (36 berths) |
| Vehicle category | M3 passenger bus (diesel, BS-VI, 12 m)                 |
| Region           | India                                                  |
| Version          | 1.0                                                    |
| Effective date   | 2026-06-01                                             |

### 1. About This Manual

This manual covers operation, passenger safety, preventive maintenance and telematics thresholds for the Coachline CX-32 12-metre intercity bus. As a public service vehicle carrying passengers, the CX-32 has stricter behaviour thresholds and additional safety equipment compared with goods vehicles.

### 2. Vehicle Specifications

Table 2.1 – CX-32 technical specifications

| Specification              | CX-32 AC Seater                            | CX-32 AC Sleeper                           |
|----------------------------|--------------------------------------------|--------------------------------------------|
| Engine                     | 5.6 L inline-6 diesel, BS-VI, rear-mounted | 5.6 L inline-6 diesel, BS-VI, rear-mounted |
| Maximum power              | 250 hp @ 2,300 rpm                         | 250 hp @ 2,300 rpm                         |
| Maximum torque             | 950 Nm @ 1,200–1,800 rpm                   | 950 Nm @ 1,200–1,800 rpm                   |
| Transmission               | 6-speed manual (6-speed AMT optional)      | 6-speed AMT                                |
| Seating / berths           | 45 passengers + driver + conductor         | 36 berths + driver + co-driver             |
| Gross vehicle weight (GVW) | 16,200 kg                                  | 16,200 kg                                  |
| Overall length             | 12.0 m                                     | 12.0 m                                     |
| Fuel tank capacity         | 350 L                                      | 350 L                                      |
| AdBlue (DEF) tank capacity | 40 L                                       | 40 L                                       |
| Speed governor setting     | 80 km/h                                    | 80 km/h                                    |
| Brakes                     | Full air disc brakes with ABS and EBD      | Full air disc brakes with ABS and EBD      |
| Auxiliary braking          | Electromagnetic retarder                   | Electromagnetic retarder                   |
| Tyre size                  | 295/80 R22.5                               | 295/80 R22.5                               |

### 3. Fluid and Consumable Capacities

**Table 3.1 – CX-32 fluid capacities**

| System                        | Capacity | Specification                                        |
|-------------------------------|----------|------------------------------------------------------|
| Engine oil (with filter)      | 22 L     | SAE 15W-40, API CK-4                                 |
| Engine coolant                | 55 L     | OAT coolant, 50:50 premix (long rear-engine circuit) |
| Transmission oil              | 9 L      | SAE 75W-90 synthetic                                 |
| Rear axle oil                 | 13 L     | SAE 85W-140, API GL-5                                |
| Power steering fluid          | 4.5 L    | ATF Dexron III                                       |
| Diesel exhaust fluid (AdBlue) | 40 L     | ISO 22241 compliant AdBlue                           |
| Fire suppression agent (FDSS) | 6 L      | Aqueous film-forming agent, sealed cylinder          |

### 4. Tyre Specifications and Pressures

**Table 4.1 – CX-32 tyre pressures**

| Position            | Cold pressure     | Minimum tread depth (company policy) |
|---------------------|-------------------|--------------------------------------|
| Front axle (single) | 7.9 bar (115 psi) | 4.0 mm                               |
| Rear axle (dual)    | 7.6 bar (110 psi) | 3.0 mm                               |

Retreaded tyres are not permitted on the front axle of any passenger bus. Check the inner dual tyres for pressure matching; a difference of more than 0.3 bar between duals causes uneven wear.

### 5. Passenger Safety Equipment

The driver must confirm during the pre-trip check that all passenger safety equipment below is present, sealed and unobstructed.

**Table 5.1 – CX-32 passenger safety equipment**

| Equipment                                    | Quantity (Seater)  | Quantity (Sleepers) | Location                      |
|----------------------------------------------|--------------------|---------------------|-------------------------------|
| Fire extinguisher, 5 kg ABC                  | 2                  | 2                   | Driver area and rear          |
| Fire detection and suppression system (FDSS) | 1                  | 1                   | Engine compartment, automatic |
| Emergency exits (rear door + roof hatches)   | 1 door + 2 hatches | 1 door + 2 hatches  | Rear right, roof              |
| Break-glass emergency windows                | 4                  | 6                   | Both sides                    |
| Emergency hammers                            | 6                  | 8                   | Next to break-glass windows   |
| ALS-140 emergency (panic) buttons            | 4                  | 6                   | Passenger area, wired to VLTD |
| First-aid kit                                | 1                  | 1                   | Driver cabin                  |
| CCTV cameras                                 | 2                  | 3                   | Front and saloon              |

### 6. Instrument Cluster Warning Lights

**Table 6.1 – CX-32 warning lamps**

| Warning lamp            | Colour | Meaning                                          | Driver action                                                           |
|-------------------------|--------|--------------------------------------------------|-------------------------------------------------------------------------|
| Engine compartment fire | Red    | Fire detected in engine compartment; suppression | Stop immediately, open all doors, evacuate passengers upwind, call 112. |

| Warning lamp                 | Colour | Meaning                                       | Driver action                                                                             |
|------------------------------|--------|-----------------------------------------------|-------------------------------------------------------------------------------------------|
| (FDSS)                       |        | discharged                                    |                                                                                           |
| Door open while moving       | Red    | A passenger door is not fully closed          | Stop and close the door; the door-brake interlock prevents moving off with the door open. |
| Low air pressure             | Red    | System air pressure below 5.5 bar             | Stop safely immediately.                                                                  |
| Engine coolant temperature   | Red    | Coolant above 110 °C                          | Stop, idle 2 minutes and switch off; check rear coolant reservoir.                        |
| Check engine (MIL)           | Amber  | Engine or emissions fault stored              | Book inspection at end of the trip.                                                       |
| AdBlue (DEF) low             | Amber  | DEF level below 10%                           | Refill AdBlue at the next stop.                                                           |
| ABS/EBD fault                | Amber  | ABS or EBD disabled                           | Drive gently; book inspection.                                                            |
| AC high-pressure fault       | Amber  | Air-conditioning compressor protection active | Switch AC to fan mode; inform depot.                                                      |
| Door-brake interlock engaged | Green  | Parking brake held because a door is open     | Information only.                                                                         |

## 7. Common Diagnostic Trouble Codes (DTC)

**Table 7.1 – Frequent CX-32 DTCs**

| DTC   | Description                                                 | Severity | Recommended action                                        |
|-------|-------------------------------------------------------------|----------|-----------------------------------------------------------|
| P0217 | Engine over-temperature condition                           | Critical | Stop safely; check coolant and rear radiator fan.         |
| P0128 | Coolant temperature below thermostat regulating temperature | Low      | Book thermostat inspection.                               |
| P0520 | Engine oil pressure sensor circuit fault                    | Medium   | Verify with the mechanical gauge; inspect sensor.         |
| P2BAD | NOx exceedence – root cause unknown                         | Medium   | Check AdBlue quality; emissions inspection.               |
| B1A50 | FDSS detection circuit fault                                | High     | Fire detection may be disabled; inspect before next trip. |
| C1F10 | Door-brake interlock switch fault                           | High     | Do not operate with passengers until repaired.            |

## 8. Preventive Maintenance Schedule

**Table 8.1 – CX-32 preventive maintenance schedule**

| Interval                | Service items                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Daily (pre-trip)        | Walk-around, tyres, lights, emergency exits and hammers, fire extinguishers, FDSS indicator, panic buttons, door interlock test |
| Weekly                  | AC return-air filter cleaning, saloon lighting, CCTV function check                                                             |
| Every 10,000 km         | Chassis greasing, brake pad wear check, air dryer check                                                                         |
| 30,000 km or 6 months   | Engine oil and oil filter change, fuel filters, air filter inspection                                                           |
| 60,000 km or 12 months  | AC system service, FDSS inspection and recertification, coolant concentration test, brake disc measurement                      |
| 120,000 km or 24 months | Coolant replacement, transmission oil replacement, axle oil replacement                                                         |

## 9. Telematics Thresholds for CX-32

Passenger buses use the strictest behaviour thresholds in the fleet because standing and sleeping passengers can be injured by forces that would be harmless in a goods vehicle.

**Table 9.1 – CX-32 telematics thresholds**

| Signal / event         | Warning threshold                 | Critical threshold                       |
|------------------------|-----------------------------------|------------------------------------------|
| Harsh braking          | Deceleration $\geq 0.25$ g        | Deceleration $\geq 0.35$ g               |
| Harsh acceleration     | Acceleration $\geq 0.20$ g        | —                                        |
| Harsh cornering        | Lateral $\geq 0.22$ g             | Lateral $\geq 0.30$ g                    |
| Overspeed              | > 80 km/h for 10 s                | > 85 km/h                                |
| Door open while moving | —                                 | Any occurrence above 5 km/h              |
| Continuous driving     | 4 h 30 min without a 30-min break | 5 h without a 30-min break               |
| Coolant temperature    | $\geq 103$ °C                     | $\geq 110$ °C                            |
| Panic button pressed   | —                                 | Any press (immediate control-room alert) |

## 10. Driver Hours and Night Operation

CX-32 night services longer than 8 hours of driving must be operated with two drivers. Drivers must change over at least every 4 hours and each driver must follow the statutory limits in the Fleet Operations Handbook (no more than 5 hours of continuous work without a 30-minute rest). The co-driver must rest in the crew berth, not in a passenger seat.

## 11. Fuel Efficiency Guidance

**Table 11.1 – CX-32 expected fuel economy**

| Variant / condition | Expected fuel economy | Notes                      |
|---------------------|-----------------------|----------------------------|
| AC Seater, highway  | 3.8–4.3 km/L          | Full passenger load, AC on |
| AC Sleeper, highway | 3.5–4.0 km/L          | Higher body weight         |
| Any variant, AC off | +10–15% better        | AC compressor load removed |

## 12. Emergency Procedures

### 12.1 Engine Compartment Fire

1. The FDSS discharges automatically and the red fire lamp comes on.
2. Stop at the nearest safe place, apply the parking brake and switch off the engine.
3. Open all doors and direct passengers out through the nearest exit, away from traffic and upwind.
4. Use the 5 kg extinguishers only if it is safe; never open the engine hatch fully on an active fire.
5. Call 112 and the depot control room. Count passengers against the manifest.

### 12.2 Passenger Evacuation

If the main door cannot be used, use the rear emergency door, then roof hatches, then break-glass windows using the emergency hammers. Assist elderly passengers, children and passengers with disabilities first. The conductor or co-driver exits last after checking the saloon and sleeper berths.

### 12.3 Panic Button Alert

When a passenger presses an AIS-140 emergency button, an alert with location is sent to the state emergency response centre and the fleet control room. The driver must not reset the alarm; the control room will call the driver to confirm the situation.

## 13. Frequently Asked Questions

Q: How often is the engine oil changed on the CX-32? A: Every 30,000 km or 6 months, whichever comes first.

Q: Why is the harsh braking threshold lower on the CX-32? A: Passengers can be injured at 0.25 g, so buses are flagged earlier than goods vehicles.

Q: How many panic buttons does the CX-32 Sleeper have? A: Six AIS-140 emergency buttons in the passenger area.